

2022 SCIENCE OLYMPIAD NATIONAL INVITATIONAL

JANUARY 8, 2022

Food Science B

Directions: Welcome to Food Science for the SO National Invitational! There will be a hands on portion for this event for the invitational. Your coaches should have a solution made for you to test your devices, and you can enter your answers at the end of the test. You do not need to wear PPE/safety equipment for the entire event, but please wear appropriate clothing and protection when doing the hands on portion. Good luck!

FROM THE HOSTS OF THE 2022 NATIONAL TOURNAMENT



**SOUTHERN CALIFORNIA
SCIENCE OLYMPIAD**

Caltech



1. (1 point) Sugar alcohols contain how many calories per gram?
 - A. 0
 - B. 0.5
 - C. 2**
 - D. 4
 - E. 10
2. (1 point) Artificial sweeteners are often _____ compared to sugar.
 - A. Much less sweet (several hundred times less sweet)
 - B. Less sweet (about 10x less sweet)
 - C. About as sweet
 - D. More sweet (up to 10x more sweet)
 - E. Much more sweet (several hundred times less sweet)**
3. (1 point) Individuals with Phneylketonuria must avoid which sweetener?
 - A. Saccharin
 - B. Sucrose
 - C. Sucralose
 - D. Aspartame**
 - E. Stevia
4. (1 point) Stevia is able to activate the sweet receptors in the tongue but cannot be metabolized by the human body, so it is a zero-calorie sweetener.
 - A. True**
 - B. False
5. (1 point) Which of these sugars has the highest relative sweetness in terms of taste?
 - A. Lactose
 - B. Maltose
 - C. Fructose**
 - D. Sucrose
 - E. Glucose
6. (1 point) What is the glycemic index of glucose?
 - A. 0
 - B. 25
 - C. 45
 - D. 65
 - E. 100 or higher**
7. (1 point) What is the glycemic index of maltose?

- A. 0
 - B. 25
 - C. 45
 - D. 65
 - E. 100 or higher**
8. (1 point) What is the glycemic index of sucrose/table sugar?
- A. 0
 - B. 25
 - C. 45
 - D. 65**
 - E. 100 or higher
9. (1 point) Sucrase breaks down sucrose into which 2 sugars?
- ☒ **Glucose**
 - ☐ Maltose
 - ☒ **Fructose**
 - ☐ Lactose
 - ☐ Galactose
10. (1 point) Pasteurization involves heating something to above 100 degrees celsius.
- A. True
 - B. False**
11. (1 point) Which of the following pathogens will pasteurization not be effective against?
- A. E. coli
 - B. E. coli O157
 - C. S. aureus
 - D. Botulinum spores**
 - E. M. tuberculosis
 - F. Salmonella
12. (1 point) Pasteurization of milk can be done in as little as how many seconds under high temperature short time (HTST) conditions?
- A. 5 seconds
 - B. 10 seconds
 - C. 15 seconds**
 - D. 60 seconds
 - E. 90 seconds
13. (2 points) Which of the following dietary fibers are soluble in water?

☒ **Pectin**

☐ Cellulose

☐ Chitin

☒ **Carrageen**

☒ **Inulin**

14. (1 point) What chemical element is the key to reactivity in a test for starches?

Iodine

15. (1 point) The test will turn what color in the presence of starch?

A. Cloudy white

B. Orange

C. Red

D. Dark blue

E. Yellow

16. (2 points) The dye used in the starch test will react with which of the following molecules?

☐ Amylopectin

☒ **Amylose**

☐ Chitin

☐ None of the above

17. (3 points) What is one non-food based use of the starch test? Why does this work? (Hint: \$)

Solution: It can be used to detect counterfeit currency. (1pt) Most currencies are printed on cotton paper or non-starch paper while counterfeits can be printed on starchy papers. (2pts)

18. (3 points) Why does the Maillard Reaction not work as well on wet foods? (ie- think of why it's recommended to pat potatoes dry prior to frying them)

Solution: The food won't get hot enough or it won't get hot enough for the reaction in a reasonable amount of cooking time. The Maillard reaction needs high temps (300 degrees F+) to occur. With high moisture content, the temperature will remain around 212F until most of the moisture boils off.

19. (2 points) You bought a bag of chips with the following nutrition label. How many calories are in a serving? (do not include units)

Total Fat 10g	16%
Saturated Fat 1g	6%
Polyunsaturated Fat 4.5g	
Monounsaturated Fat 4.5g	
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 180mg	7%
Potassium 330mg	9%
Total Carbohydrate 15g	5%
Dietary Fiber 1g	4%
Sugars 0g	
Protein 2g	

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20. (3 points) How are serving sizes determined and updated?

Solution: They are determined and updated depending on how much people realistically eat in one setting. In containers like beverage bottles or single serving chip bags, serving size can reflect the contents of the entire package.

21. (2 points) The new nutrient labeling guidelines require manufacturers to list the % daily value and actual amounts for which of the following vitamins?

- ☐ Vitamin A
- ☐ Vitamin B6
- ☐ Vitamin C
- ☒ **Vitamin D**
- ☐ Vitamin E
- ☐ Vitamin K

22. (2 points) The recommended daily values of which of the following nutrients have been updated in the new nutrition labels as a result of new evidence?

- ☐ Iron
- ☒ **Sodium**
- ☒ **Vitamin D**
- ☐ Cholesterol
- ☒ **Dietary fiber**
- ☐

23. (2 points) How does smoking salmon preserve it?

Solution: The chemicals from wood smoke prevent bacterial growth/prevent bacteria from entering the flesh of the fish

24. (3 points) How is a hydrometer different from a hygrometer?

Solution: Hydrometer measures amount of dissolved solutes (like salt, sugar, etc.) in a solution while a hygrometer measures relative humidity

25. (1 point) How many carbons are lost as carbon dioxide per pyruvate in the citric acid cycle?

- A. 0
- B. 1
- C. 2**
- D. 3
- E. 4

26. (1 point) Is the citric acid cycle oxidative or reductive?

- A. Oxidative**
- B. Reductive
- C. Both- it depends on the step
- D. Neither

27. (1 point) Curdling of milk is due to the aggregation of what protein?

Casein

28. (1 point) Milk can curdle in warm/hot temperatures and when it goes bad. Which of the following methods is also a common way of causing milk to curdle?

- A. Addition of acid**
- B. Addition of base
- C. There are no other ways to curdle milk
- D. Freezing milk

29. (2 points) Which of the following allergens are the most common cause of allergic reactions in children? Choose the top 3 answers.

- ☐ Wheat
- ☒ **Milk**
- ☐ Shellfish
- ☐ Soy
- ☒ **Peanuts**
- ☒ **Eggs**

30. (2 points) Jane and Joe both do not like milk. If Jane were to consume cow's milk, she would only experience mild gastrointestinal discomfort; whereas Joe will develop hives, need medical assistance and can potentially die from consuming milk. What are Jane and Joe's medical relationships with milk?

Solution: Jane has lactose intolerance Joe has a milk allergy

31. (2 points) Sucralose is actually just a modified form of sucrose. What is the chemical formula of sucralose? What process is used to transform sucrose to sucralose?

Solution: C₁₂H₁₉Cl₃O₈ Chlorination of sucrose

32. (1 point) What is the sweetness of sucralose relative to sucrose? Pick the answer that is closest.

- A. 10x more
- B. 10x less
- C. 100x more
- D. 100x less
- E. 500x more**
- F. 500x less

33. (1 point) How does crème brulee get its characteristic hardened sugar topping?

Solution: Sugar on the top gets caramelized

34. (2 points) Jonathan is a gastronomist-in-training. His favorite aspect about cooking is searing steaks. Jonathan notices that there is a color change in the steak as he cooks it from rare to medium well. What causes the color change and at what temperature should this color change start?

Solution: Myoglobin denaturation. Should occur at 140F

35. (2 points) Jonathan also notices that in addition to the color change, the steak slowly becomes more opaque. What is causing this change in opacity?

Solution: The protein molecules have denatured and unfolded. They coagulate, making it impossible for light to pass through.

36. (1 point) Jonathan has been told that dry, overcooked steaks do not lack water inside. Rather, he was told that their toughness is due to a specific denatured protein squeezing out the liquid in the muscle fibers. What is this specific protein?

Solution: Actin

Karen has picked up the quarantine hobby of kombucha making. She adds sucrose and black tea leaves to water, and then adds her symbiotic culture of bacteria and yeast (SCOBY) to a jar. She then sets the jar in her closet at room temperature (75F) for 2 weeks.

37. (1 point) What is name of the commonly used food preservation technique that Karen is using to make her kombucha?

Solution: Fermentation

38. (3 points) Similar to how beer and vinegar are made, the SCOBY metabolizes the sucrose using the process in (a). What are the three major products of the chemical process?

Solution: Ethanol, CO₂, acetic acid (will also accept gluconic acid or glucuronic acid)

39. (2 points) Karen was craving kombucha during the hot July summer, so she prepared to make a giant

jar of kombucha. However, she accidentally left it out on her kitchen countertop in the sun over the weekend. She moved it into the closet but by 2 weeks it was clear that this batch was failed kombucha. What likely went wrong?

Solution: Got too hot, killed the SCOBY

40. (2 points) Karen wondered if she could make her kombucha even healthier by using Stevia instead of sucrose. She asked her friend, a seasoned kombucha enthusiast, and was told it wouldn't work. Why?

Solution: Stevia is a non-fermentable sugar, the SCOBY needs to be fed a fermentable sugar

41. (3 points) Explain, at the biological and molecular level, how yeast is able to raise bread.

Solution: Yeast ferments sugars (or carbohydrates) in the dough, producing alcohol and carbon dioxide. The carbon dioxide is responsible for raising the dough.

42. (2 points) What additional products make sourdough taste sour?

Solution: Lactic acid and acetic acid

43. (3 points) Baking powder and baking soda are two commonly used chemical leavening agents. How do they work to raise bread? What is the main functional difference between the two?

Solution: Both rely on chemical reactions between an acid and base to produce carbon dioxide gas, which raises dough. Baking soda requires addition of an acid in the recipe. Baking powder contains both base and acid components, and thus only requires moisture and heat to activate.

44. (3 points) Puff pastry does not use any of the above leavening agents. How is puff pastry made and leavened?

Solution: Puff pastry is made by layering butter and dough. When baked, the water in the butter turns into steam, expanding and raising the pastry.

45. (3 points) Crystallization consists of four primary steps. The first step is generating a supersaturated state. What are the next three steps (in chronological order)?

Nucleation

46. (3 points) Jonathan tried to make ice cream this past weekend. He made his mixture and then put it in his regular house freezer. The next day, he took out his ice cream and took a bite. He was so upset! The ice cream was icy! It tasted like it had freezer burn. What gave the ice cream that texture? What could Jonathan do next time in order to minimize the nasty texture?

Solution: 1pt for Large ice crystals in the ice cream. 2pt for Jonathan needs to generate smaller crystals during the processing.

47. (10 points) Did you build a hydrometer?

A. Yes

B. No

48. (10 points) What is the percent sucrose in solution 1?

Solution: 3% (Full 10 points if between 2-4%; 7 points if 1-5%; 4 points if 0-6%)

49. (10 points) What is the percent sucrose in solution 2?

Solution: 5% (give full 10 points if 4-6%; give 7 points if 3-7%; give 4 points if 2-8%)